Grinding at the Edge: Empirical Signer-Cost Distributions for Bitcoin-Tuned Hash-Based Signatures

George Nanos · DRAFT 2026-10-24 · under review; open items are listed in §7

> **Draft status.** Every number in this draft comes from `analysis/out/` (run `196e7e91e907`) or from `kb/claims/all_claims.yaml`. Signature sizes use R = 16 B (FIPS 205, SPHINCS-Parameters HEAD); Kudinov–Nick's tables use R = 32 B and are 16 B larger.

Abstract

Hash-based signatures are the conservative post-quantum option for Bitcoin. Their size can be cut by *grinding*: the signer retries a hash until a condition holds, so checksum chains (WOTS+C) or FORS tree paths (FORS+C, PORS+FP) can be dropped from the signature. Prior work tabulates the expected search cost and an analytic 2^{-30} worst case in abstract hash calls.

We measure the cost. On a 45 W laptop GPU we ran 11.16 million reference-verified grinds across 37 grind configurations, which serve 40 parameter sets at 2^{20}, 2^{30} and 2^{40} signature limits. We cross-checked them on a SHA-NI laptop CPU. Results:

- Every measured mean is within 2 % of the analytic expectation, and the distributions are geometric down to the 10^{-4}…10^{-6} tail.
- The widely used cost model under-prices FORS+C and PORS+FP attempts by 1.5–1.75× under FIPS 205's SHA-2 instantiation, and over-prices the counter-in-H_msg variant by 1.33×.
- Grinding is at most 18 % of total signing work; tree building dominates.
- A signer budget of 10 s at p99 on a laptop CPU, or under one second on a laptop GPU, buys 596–1,052 bytes per signature over a hardware-wallet-safe set: 17–23 % of a 1-input/2-output transaction.
- Verification cost never increases.

We release the CUDA/NVRTC grinder, the dataset and a one-command reproduction.

1. Introduction

BIP-360 (P2MR) was merged on 2026-02-11 and leaves the post-quantum signature opcode to a future BIP. Kudinov and Nick's Bitcoin-tuned SPHINCS+ variants (ePrint 2025/2203) show that two levers shrink hash-based signatures:

- **fewer signatures per key:** 2^{20}–2^{40} instead of 2^{64};
- **grinding:** a pure work-for-bytes trade made at signing time.

How much work a signer can afford is a free parameter of the design, and the community disagrees about it. Positions range from "seconds per input on low-power devices is fine" (Maxwell, RQ-02) to "a couple of minutes is not a big deal" (Blockstream, RQ-05). SHRINCS fixes a single parameter set (RQ-41).

Kudinov and Nick opened the bitcoindev thread with the question **"What are the concrete performance requirements across various hardware, including low-power devices?"** (OQ-1). Maxwell's answer to whether several parameter sets should be standardized was **"it would depend on their relative performance"** (OQ-2).

This paper supplies that relative performance, measured. Contributions:

1. **Validation (RQ1).** The analytic expected cost holds to 2 % for WOTS+C, FORS+C (two layouts) and PORS+FP, and the full distribution is geometric.
2. **Per-attempt pricing (H1b).** Counted under FIPS 205 SHA-2, grinding attempts cost 1 / 6 / 3 / 7 SHA-256 compressions (WOTS+C / FORS+C R-grind / FORS+C ctr-grind / PORS+FP), not 1 / 4 / 4 / 4.
3. **Latency per signer class (RQ2, OQ-1).** p50 / p99 / p99.9 signing time from hardware wallets to GPUs, measured on two classes and scaled from cited rates for four.
4. **Frontier and fees (RQ3/RQ4, OQ-2).** The smallest signature per time budget and per class, and its fee effect.
5. **Open artifacts:** kernels, data, notebooks, reproduction.

2. Background

SPHINCS+ / SLH-DSA. A hypertree of d layers of XMSS trees, each signing the root below with WOTS, over a few-time signature (FORS) on the message digest. FIPS 205 fixes the SHA-2 instantiation:

- H_msg = MGF1-SHA-256(R ‖ PK.seed ‖ SHA-256(R ‖ PK.seed ‖ PK.root ‖ M), m);
- PRF_msg = HMAC-SHA-256.

WOTS+C. The signer grinds a counter until the base-w digits of d = Th(PK.seed, ADRS, m ‖ count) sum to a fixed target S. This removes the checksum chains. Success probability p = v / w^len1. The signature carries the counter.

FORS+C. The signer grinds until the last FORS tree opens leaf 0, which happens with $p = 2^{-a}$, so that tree's path is dropped. Kudinov–Nick put the counter into PRF_msg ("R-grind": fresh R per attempt, nothing extra stored). The original SPHINCS+C appends a counter to H_msg ("ctr-grind": R fixed, 4-byte counter stored).

PORS+FP. PORS with forced pruning. The signer grinds R until the k distinct indices derived from H_msg have an Octopus authentication set of size ≤ m_max. Expected work comes from Abri–Katz's exact PMF. With a 512-bit H_msg, an attempt can also fail to yield k distinct indices; we include that probability (P_enough, 0.987–1.000 for the reference rows).

Tail model. Independent attempts make the number of tries T geometric: P(T > t) = (1 − p)^t. So p99 ≈ 4.61/p, p99.9 ≈ 6.91/p and the 2^{-30} quantile ≈ 20.8/p.

3. Methodology

Parameter points. We use 40 shortlisted points:

- the 29 Kudinov–Nick table rows (reproduced exactly by our port of `costs.sage`: size, verification cost, expected search and 2^{-30} worst case);
- per signature limit, a **hw_wallet_safe** point (every grind ≤ 2^{12} tries), a **gpu_signer_moderate** point and a **gpu_signer_max** point;
- two stress points.

The search space was capped at grinds ≤ 2^{24} expected tries and signing ≤ 2^{30} compressions. The grind distribution depends only on (w, S, z), on (k, a) and the layout, or on (k, a, h, m_max), so the 40 points need 37 grind configurations.

Byte layouts. The authors give no byte-level SHA-256 spec for FORS+C or PORS+FP, so we fixed FIPS 205-faithful layouts (`kb/constructions/`):

- WOTS+C: ADRSc (type 7) ‖ m ‖ count_{32} in one block after the PK.seed midstate;
- FORS+C: last a-bit chunk of md is zero;
- R-grind: HMAC message opt_rand ‖ M ‖ i_{32}.

Tries do not depend on these choices; per-attempt cost and test vectors do.

Kernels. NVRTC-compiled CUDA (no toolkit needed), one kernel per layout. Each thread strides counters, and the lowest successful counter wins (atomicMin), so the result is deterministic and equals a sequential search. Many independent instances are batched per launch, and each instance is one sample.

Correctness.

- **Gate.** A campaign starts only after the E1 gate passes: 10 configurations at reduced difficulty (all acceptance paths, including FORS masks straddling 32-bit words and non-power-of-two PORS t), 10^4 instances each. Every counter matches the pure-Python reference with a full minimality check.
- **Campaign.** All 11.16 M campaign counters were re-checked by the reference.
- **CPU cross-check.** A separate SHA-NI CPU grinder reproduced all WOTS+C counters bit for bit.

Samples. 10^6 per WOTS+C config, $2{\cdot}10^5$ per FORS+C config, 10^5 per PORS+FP config. The three PORS+FP configs with E[tries] ≈ $1.2{\cdot}10^7$ got $2{\cdot}10^4$.

Hardware. MSI Thin 15 B13VE:

- RTX 4050 Laptop GPU at its 45 W TGP floor (20 SMs, sm_89). E0 sustained single-block SHA-256: $2.15{\cdot}10^9$ compressions/s.
- i5-13420H (4P+4E, SHA-NI). E5 sustained, 12 threads: $2.33{\cdot}10^8$ compressions/s; 1 thread: $2.93{\cdot}10^7$.

Board power, clocks and clock-event reasons were logged at 10 Hz. The GPU sat at its power cap; 0.014 % of busy samples were flagged by any slowdown reason.

Signer classes. For the other classes we use cited typical rates (`kb/hardware.yaml`), labelled as cited/derived:

- Cortex-M4 software SHA: $4{\cdot}10^4$/s;
- MCU with SHA peripheral: $1.2{\cdot}10^5$/s;
- Raspberry Pi 5: $7.4{\cdot}10^7$/s;

- 8× RTX 4090: $1.72 \cdot 10^{11}$/s.

Full signing cost (E4) = the model's non-grinding work (tree building) + the measured per-signature grind distribution (d WOTS+C layers + one FTS grind, resampled), priced at the counted per-attempt cost.

Statistics. Mean ratio with 95 % CI. A randomized probability integral transform against the *analytic* p (nothing fitted), tested with KS (Holm-corrected) and Anderson–Darling. Σz^2 across configs.

4. Results

4.1 Validation (RQ1, H1)

	value
configs / samples	37 / 11.16 M, all reference-verified
max |mean/analytic − 1|	1.66 % (`pors-k9-a13-h48-m77`, n = $2 \cdot 10^4$, z = −2.36); rerun at n = 10^5: 0.08 %, z ≈ −0.26
Σz^2 (37 dof)	39.6, p = 0.36; 2 configs outside their 95 % CI (1.9 expected)
KS on PIT, Holm 5 %	0 rejections
AD on PIT, 5 %	2 rejections (1.9 expected)
variance / geometric variance	0.963–1.026 (median 1.000)
p99 / mean	4.56–4.65 (theory 4.61)
p99.9 / mean	6.73–7.05 (theory 6.91)

All configs collapse onto e^(−x) when tries are divided by E[tries] (Fig. 1). The analytic model, including the P_enough correction for PORS+FP, is accurate for the mean and the shape.

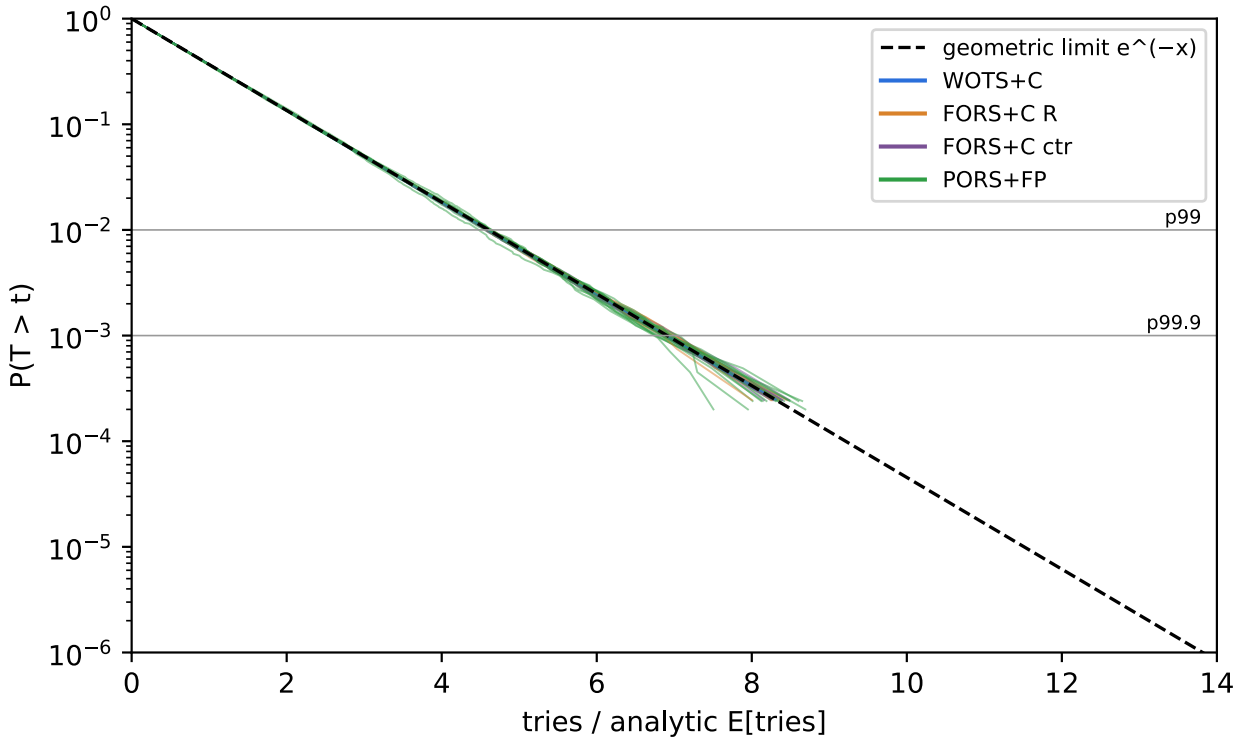


Figure 1. Survival function of tries divided by E[tries] for all 37 configurations; every curve follows e^(−x).

4.2 What an attempt really costs (H1b)

grind	model (costs.sage)	counted (FIPS 205 SHA-2)	GPU attempts/s	useful comp/s ÷ E0
WOTS+C	1	1	$1.35 \cdot 10^9$	0.63
FORS+C R-grind	4	6	$3.49 \cdot 10^8$	0.97
FORS+C ctr-grind	4	3	$7.25 \cdot 10^8$	1.01
PORS+FP	4	7	$1.81 \cdot 10^8$	0.59

The model assumes a single-SHA-256 H_msg. FIPS 205's H_msg is nested (inner hash + MGF1) and PRF_msg is HMAC. Two consequences:

- **Layout choice.** On the GPU, ctr-grind runs 2.08× more attempts per second than R-grind but stores a 4-byte counter. This is a measurable design choice, not just a 4-byte question.
- **Kernel efficiency.** WOTS+C and PORS+FP kernels run at about 60 % of raw SHA throughput because of per-attempt digit and Octopus work. FORS+C runs at raw speed.

Re-priced full signing cost differs from the model by −1 % … +8 %.

4.3 Grinding is a minor part of signing

Mean grinding is 0–18 % of total signing work across all 49 point × layout rows; tree building dominates. Even for the gpu_signer_max points (≈ $5–8 \cdot 10^8$ compressions), the base cost is 86–90 % of the total. Two consequences:

- the grind tail moves the full-signing p99 only slightly;

- the cost of the small "GPU-signer" sets is mainly larger hypertree layers (h/d = 16), not grinding.

Per signature, p99.9/mean of the grind ranges from 2.96 (several comparable WOTS+C layers, which concentrate) to 7.1 (one dominant FTS grind). Kudinov–Nick's 2^{-30} worst case is 6.3–20.9× the mean.

4.4 Latency per signer class (RQ2, OQ-1)

p99 full signing time = model base + measured grind, divided by the class rate:

limit	profile	bytes	Cortex-M4 [c]	MCU+SHA [c]	Pi 5 [c]	i5 SHA-NI [m]	RTX 4050 [m]
2^{20}	hw_wallet_safe	2876	2.7 min	54 s	88 ms	28 ms	3.0 ms
2^{20}	gpu_signer_moderate	2504	24 min	8.0 min	0.78 s	0.25 s	27 ms
2^{20}	gpu_signer_max	2280	6.1 h	2.0 h	12 s	3.8 s	0.41 s
2^{30}	hw_wallet_safe	3456	2.2 min	43 s	70 ms	22 ms	2.4 ms
2^{30}	gpu_signer_moderate	2908	20 min	6.7 min	0.65 s	0.21 s	22 ms
2^{30}	gpu_signer_max	2664	6.3 h	2.1 h	12 s	3.9 s	0.42 s
2^{40}	hw_wallet_safe	3944	2.8 min	56 s	90 ms	29 ms	3.1 ms
2^{40}	gpu_signer_moderate	3216	35 min	12 min	1.1 s	0.36 s	39 ms
2^{40}	gpu_signer_max	2892	8.0 h	2.7 h	16 s	5.0 s	0.54 s

[c] cited/derived rate · [m] measured on the test laptop. Times assume perfect parallelism within a class (§7).

4.5 Frontier (RQ3, H2, OQ-2)

Fig. 2 plots bytes against p99 signing work per limit. Smallest shortlisted signature within a p99 budget:

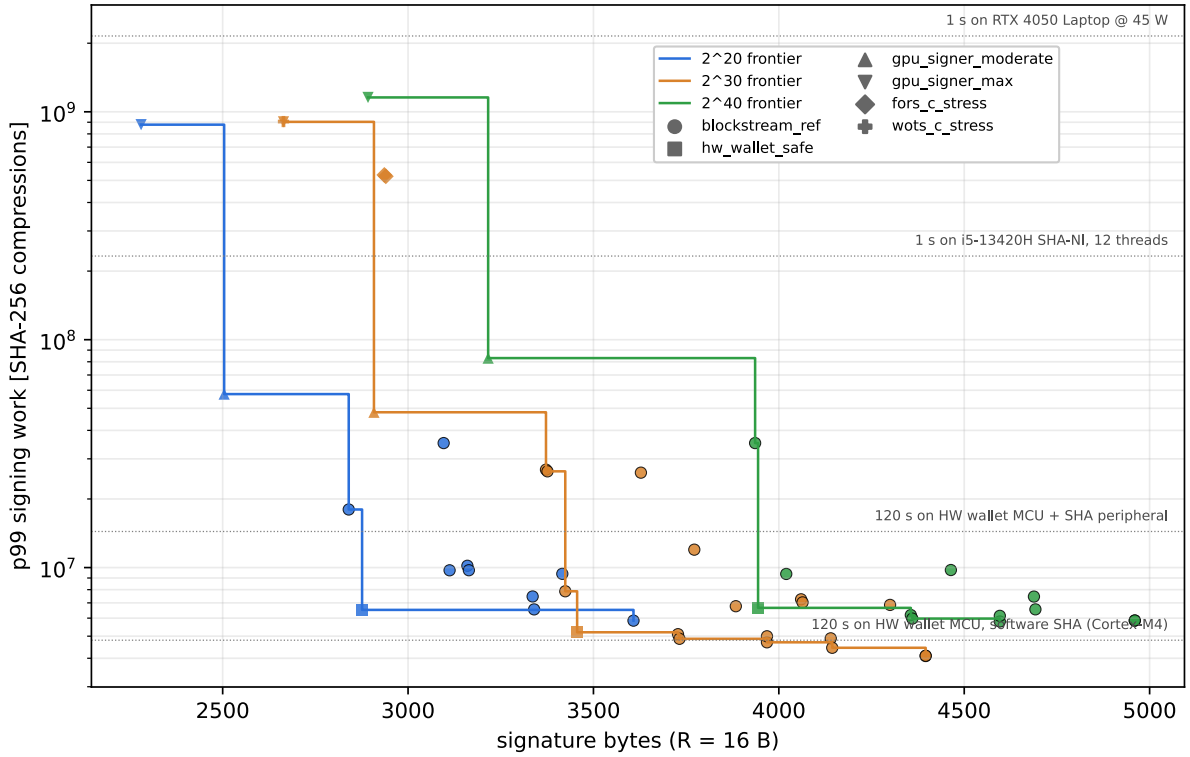


Figure 2. Signature bytes against p99 signing work (SHA-256 compressions) per signature limit, with the Pareto frontier and one-second and two-minute budgets of four signer classes.

limit	Cortex-M4, 120 s	MCU+SHA, 120 s	laptop CPU, 1 s	laptop CPU, 10 s	laptop GPU, 1 s
2^{20}	none ≤ 120 s	2876	2504	2280	2280
2^{30}	3968	3424	2908	2664	2664
2^{40}	none ≤ 120 s	3944	3216	2892	2892

The GPU reaches the smallest shortlisted point at every budget. Because the shortlist caps signing at 2^{30} compressions, the true GPU frontier lies further left, and the GPU savings are lower bounds.

4.6 Fees (RQ4)

Per input, against the hw_wallet_safe point of the same limit (BIP-360 P2MR, depth 1, 32-byte PQ key):

limit	profile	Δbytes	ΔvB	Δsat @ 2 / 10 / 50 sat/vB	share of a 1-in/2-out tx
2^{20}	gpu_signer_moderate	−372	−93	−186 / −930 / −4,650	−10.6 %
2^{20}	gpu_signer_max	−596	−149	−298 / −1,490 / −7,450	−17.0 %
2^{30}	gpu_signer_moderate	−548	−137	−274 / −1,370 / −6,850	−13.4 %
2^{30}	gpu_signer_max	−792	−198	−396 / −1,980 / −9,900	−19.4 %
2^{40}	gpu_signer_moderate	−728	−182	−364 / −1,820 / −9,100	−15.9 %
2^{40}	gpu_signer_max	−1,052	−263	−526 / −2,630 / −13,150	−23.0 %

4.7 Verification (H3) and energy (RQ5)

- **Verification.** In all 15 pairs that differ only in FTS compression, verification compressions fall by 15–40. The gpu_signer points verify cheaper than the hw_wallet_safe points of the same limit (e.g. 2^{40}: 5,893 vs. 10,026 compressions) because they have fewer, taller layers.
- **Energy.** Measured GPU energy is 20.8 nJ per raw compression. Per attempt: WOTS+C 27 nJ, ctr-grind 59 nJ, R-grind 124 nJ, PORS+FP 241 nJ. The largest GPU-signer signature costs about 18 J on the RTX 4050. Energy is not a constraint for desktop or GPU signers.

5. Discussion

Design for 7× the mean, not the mean. The tails are textbook geometric. A wallet UI should budget about 7× the mean grind (p99.9) and expose progress, not a fixed timeout at the mean. Because tree building dominates, full signing time is much tighter than the grind tail; per-signature p99.9/mean of grinding alone is 3–7.

Two signer profiles are a real choice. Between a hardware-wallet-safe and a GPU-signer set the gap is 596–1,052 B (17–23 % of a simple spend) at the limits studied. Even a laptop CPU at a 1 s budget gets 372–728 B of it. Maxwell's condition for more than one parameter set ("relative performance") is therefore met with a large margin. Free configuration hurts privacy (RQ-40), which argues for exactly two named profiles rather than a menu.

Price attempts at FIPS 205 cost. Cost tables that feed parameter selection should use 6 (R-grind) or 7 (PORS+FP) compressions per attempt, or state the non-FIPS H_msg they assume. For signers that store a counter anyway, ctr-grind halves FORS+C grinding.

Overgrinding (OQ-3). Raising the WOTS+C target S trades signer work for verifier work at fixed size. At w = 4, S = 96 → 128 removes a third of per-layer chain steps for about 650× more grinding. Measured (run `oq3-eb65c6cd81a0`, 21 configs, 10^5–10^6 samples each, notebook 05, Fig. 3): every mean is within 0.26 % of analytic.

- **w = 4.** S = 128 costs $1.5{\cdot}10^4$ tries/layer, a p99 of 1.7 s on a software-SHA Cortex-M4. S = 140 (−46 % chain steps) costs $6.2{\cdot}10^6$ tries, a p99 of 12 min on that MCU but 0.12 s on a 12-thread laptop CPU.
- **w = 16.** S = 240 → 304 removes 27 % of chain steps for 20× more tries, raising the MCU p99 from 7.5 ms to 0.15 s.

Overgrinding is cheap when the signer is a PC. For a software-SHA hardware wallet at w = 4, the p99 passes 1 s once about a third of the chain steps are removed (S ≥ 128).

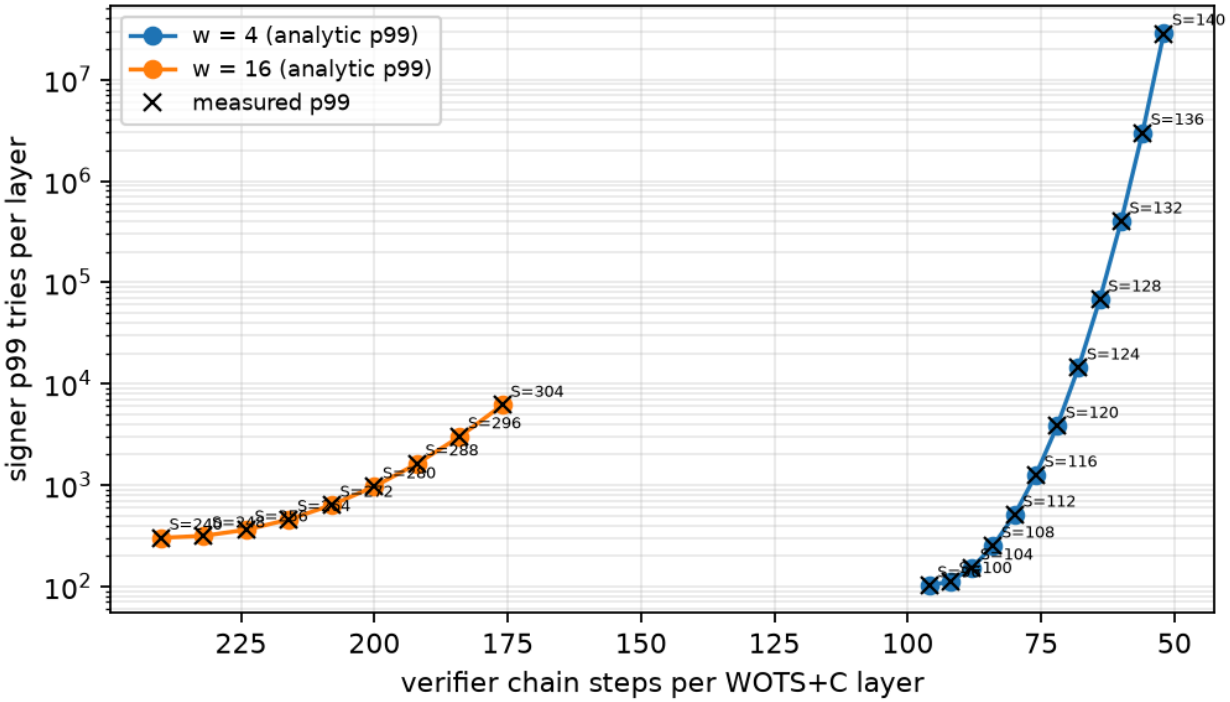


Figure 3. OQ-3 overgrinding: signer p99 tries per layer against verifier chain steps, analytic (lines) and measured (crosses).

6. Related work

- Kudinov–Nick 2025 (ePrint 2025/2203): parameters, analytic expected and 2^{-30} costs; our baseline.
- Hülsing–Kudinov–Ronen–Yogev, SPHINCS+C: WOTS+C, FORS+C.
- Abri–Katz: PORS+FP.
- Fluhrer–Dang 2024/018: smaller SPHINCS+ without grinding; grinding buys ≈ 300 B at equal signing budget.
- The SHRINCS BIP draft: stateful + plain-SLH-DSA stateless path.
- Blockstream's 2026 hardware-wallet benchmarks: means on real devices; our tails complement them.

7. Threats to validity

Summarized from `kb/threats_to_validity.md`:

- **z = 1 security (open).** The z = 1 (forced zero chain) gpu_signer points have no analysed security impact in our model. They are reported as measured, not endorsed. SPHINCS+C's theorems are stated for general z with no z-dependent bound term. What remains is independent review and the verifier rule that the zeroed digit must be checked (threats §1.2).
- **Our byte layouts.** Tries transfer to any author layout; per-attempt cost is reported for both readings of H_msg.
- **Modelled tree-building cost.** The base cost is modelled, not measured.
- **Cited ladder.** Four of six signer classes use cited rates; tries remain the primary, hardware-independent metric.

- **Parallelism assumption.** Throughput is assumed to equal latency within a class.
- **Small n.** Three configs have n = $2 \cdot 10^4$. The one with the largest deviation was rerun at 10^5 and matches the analytic mean (ratio 0.9992).
- **Tail extrapolation.** The 2^{-30} tail is extrapolated from the geometric form.
- **Hardware.** One laptop, one run.
- **Shortlist cap.** The cap makes GPU savings a lower bound.

8. Conclusion and artifacts

Grinding behaves exactly as modelled in tries, costs more per attempt than modelled under FIPS 205, and is a minor part of signing. The signer's hardware budget is a first-order size lever: hundreds of bytes per input. The PQ-signature BIP should state its signer assumption explicitly and consider two named parameter profiles.

Artifacts. `github.com/nanos007/grindstone-pq`: NVRTC kernels, SHA-NI grinder, Python reference, campaign runner, notebooks, `python -m runner.reproduce`. Data under CC-BY-4.0, dataset SHA-256 `01e9fd2ca78f89e5c2525072705e8f32f42863054ffcbc7301341905bfe98d80`.

References

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